Residual-Based Weighting. We compute importance weights based on residual magnitudes using a sigmoid transformation:

$$w_{\text{residual}}(r_t) = 0.5 + \frac{1.0}{1 + \exp(-|r_t|/(\tau \cdot \sigma_r))}, \tag{6}$$

where σ_r is the residual standard deviation and τ controls deviation sensitivity, emphasizing large residual magnitudes while maintaining stability.

Position-Based Weighting. We incorporate positional information through adaptive weighting patterns that exploit the natural information gradient in sequences. For the linear decay pattern (which achieved optimal performance), we assign higher importance to tokens at the beginning of sequences based on the intuition that early tokens establish distinctive membership signals:

$$w_{\text{position}}(t) = 1.5 - \frac{t}{T}, \tag{7}$$

where T is the sequence length. This reflects the intuition that earlier tokens in training sequences may contain more distinctive membership signals.

Multi-Scale Deviation Analysis. To capture patterns at different temporal scales and enhance robustness, we compute EMA trends using multiple smoothing factors $\{\alpha_1, \alpha_2, \alpha_3\}$ and identify tokens that consistently deviate across scales, reducing sensitivity to spurious single-scale outliers:

In the generated code, multi-scale deviation is implemented as an optional component and is not actually utilized for this experiment.

$$w_{\text{multiscale}}(t) = \prod_{i=1}^{3} \max\left(1.0, 1.0 + 0.3 \cdot \frac{|r_t^{(i)}|}{\sigma_{r_i}}\right), \tag{8}$$

where $r_t^{(i)}$ represents residuals computed with smoothing factor α_i.

3.3.2 Final Score Computation

Our enhanced membership score combines all weighting components:

$$w_t = w_{\text{residual}}(r_t) \cdot w_{\text{position}}(t) \cdot w_{\text{multiscale}}(t), \tag{9}$$

$$\text{Score}_{\text{enhanced}} = \frac{\sum_{t \in \text{top-}k\%} s_t \cdot w_t}{\sum_{t \in \text{top-}k\%} w_t}, \tag{10}$$

where the top-$k\%$ tokens are selected based on the original Min-K%++ scores but weighted according to our enhanced scheme.

3.4 Implementation Details

Our implementation builds upon the original Min-K%++ codebase, computing base normalized scores identically for fair comparison. Key hyperparameters: EMA smoothing $\alpha = 0.3$, multi-scale analysis $\{\alpha_1 = 0.1, \alpha_2 = 0.3, \alpha_3 = 0.5\}$, temperature $\tau = 2.0$, and linear decay position weighting. Computational overhead is minimal as operations are lightweight token-level computations scaling linearly with sequence length.

multi-scale analysis is not actually utilized for this experiment.

4 Experimental Setup

We evaluate our proposed method on the WikiMIA benchmark (Shi et al., 2024), a comprehensive dataset for assessing membership inference attacks. Our experimental setup provides thorough evaluation across different model architectures and sequence lengths.

Dataset. WikiMIA contains Wikipedia text excerpts for membership inference evaluation. We experiment with sequence lengths of 32, 64, and 128 tokens to analyze how distributional patterns emerge at different scales.

Model Architectures. We evaluate on two representative architectures: **Pythia-2.8b** (Biderman et al., 2023), a transformer-based model trained on the Pile dataset, and **Mamba-1.4b** (Gu & Dao, 2023), a state-space model with selective mechanisms. These architectures assess generalizability across different model paradigms.

Evaluation Metrics. We employ three standard metrics for membership inference evaluation:

- **AUROC**: Area Under the Receiver Operating Characteristic curve, measuring the overall ranking quality across all possible thresholds.
- **FPR95**: False Positive Rate at 95% True Positive Rate, indicating the method's specificity at high sensitivity operating points.
- **TPR@5%FPR** (also denoted TPR05): True Positive Rate at 5% False Positive Rate, measuring precision at low false positive rates, which is crucial for practical deployment scenarios.

Implementation Details. Our implementation builds upon the original Min-K%++ codebase to ensure fair comparison. We maintain identical tokenization, vocabulary handling, and score normalization. Key hyperparameters include: (1) EMA smoothing factor $\alpha = 0.3$, (2) temperature parameter $\tau = 2.0$ for residual weighting, and (3) linear decay position weighting with $w_{\text{position}}(t) = 1.5 - t/T$. All experiments use the same environment and random seeds for reproducibility.

5 Experiments

We present experimental results demonstrating the effectiveness of our residual score decomposition approach across different model architectures and sequence lengths. Our experiments show consistent improvements over the Min-K%++ baseline, with particularly strong gains for longer sequences.

5.1 Main Results

Our experiments demonstrate consistent improvements over the Min-K%++ baseline across all tested configurations. Figure 1 presents the most compelling evidence of our method's effectiveness on Mamba-1.4b with 128-token sequences, where distributional improvements are most pronounced. Table 1 provides comprehensive quantitative results across all model and sequence length configurations.

Consistent AUROC Improvements. Our method achieves consistent AUROC improvements across all configurations, ranging from 0.6 to 1.6 percentage points. The largest improvement occurs for Mamba-1.4b on 128-token sequences (Figure 1), where we achieve 70.0% AUROC compared to the 68.4% baseline, representing a substantial 1.6 percentage point gain. This significant improvement is accompanied by dramatic distributional changes visible in the score histograms, where our method creates more concentrated training distributions while preserving the broader, heavier-tailed patterns characteristic of non-training data.

The difference is not dramatic, and it is hard to see from the figure alone.

Enhanced Low-FPR Performance. Our method demonstrates particular strength in low false positive rate scenarios, as evidenced by improvements in TPR@5%FPR across most configurations. This enhanced precision is particularly valuable for practical deployment scenarios where false positives must be minimized. The improvements are most pronounced for configurations where position weighting can effectively emphasize the distinctive patterns in early tokens.

Model Architecture Generalization. The consistent improvements across both transformer-based (Pythia) and state-space (Mamba) architectures demonstrate that our approach captures fundamental distributional patterns that transcend specific model paradigms. Figure 2 further illustrates the robustness of our method through comprehensive hyperparameter sensitivity analysis, revealing critical performance trade-offs that guide practical deployment decisions.

Table 1: Performance comparison across models and sequence lengths. Best results are shown in **bold**. Our method achieves consistent AUROC improvements ranging from 0.6 to 1.6 percentage points across all configurations.

Model	Length	Method	AUROC	TPR@5%FPR
Pythia-2.8b	32	Min-K%++	64.4%	12.4%
		Ours	**65.0%**	**14.0%**
	64	Min-K%++	63.8%	14.1%
		Ours	**65.0%**	**14.4%**
	128	Min-K%++	66.4%	12.9%
		Ours	**67.1%**	12.9%
Mamba-1.4b	32	Min-K%++	66.8%	12.1%
		Ours	**67.8%**	**14.2%**
	64	Min-K%++	66.4%	16.5%
		Ours	**67.6%**	13.4%
	128	Min-K%++	68.4%	10.1%
		Ours	**70.0%**	**13.7%**

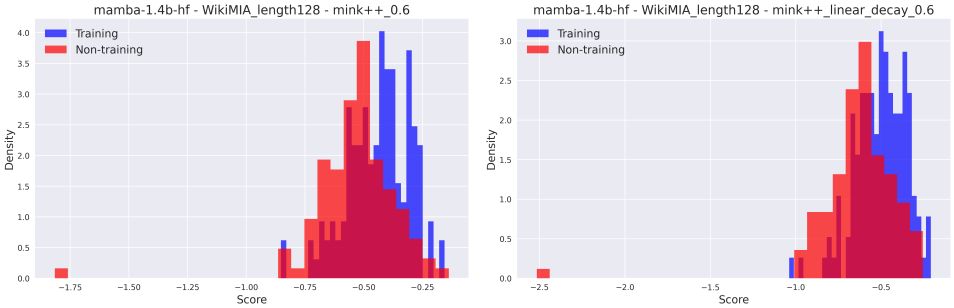

Figure 1: Score distributions for Mamba-1.4b on 128-token sequences comparing Min-K%++ baseline (left) with our proposed method (right). Training data is shown in blue and non-training data in red. The proposed method achieves superior distributional separation, with training data exhibiting more concentrated distributions while non-training data maintains broader, heavier-tailed patterns. This improved separation directly translates to the 1.6 percentage point AUROC improvement shown in Table 1. The transformation demonstrates how position-dependent weighting fundamentally alters score distribution characteristics, creating more discriminative patterns for membership detection.

5.2 Distributional Analysis

The distributional improvements provide crucial insights into why position-dependent weighting enhances membership detection. Figure 1 demonstrates that our approach fundamentally alters score distribution characteristics, creating more pronounced separation between training and non-training patterns.

Training Data Concentration. Training sequences exhibit more concentrated distributions with reduced variance. Our linear decay weighting amplifies this concentration by emphasizing early tokens with stronger membership signals, leading to tighter distributions with reduced overlap with non-training patterns.

Non-Training Data Tail Behavior. Non-training data maintains broader distributions with heavier tails, indicating higher uncertainty. Our method preserves these tail characteristics while enhancing separation from training distributions, preventing over-smoothing that could reduce discriminative power.

Sequence Length Effects. The magnitude of distributional improvements scales with sequence length, supporting our hypothesis that position-dependent patterns become more pronounced in longer